

Automatic Calibration for Inertial Measurement Unit

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Abstract—The nine degrees-of-freedom (DOF) inertial measurement units (IMU) are generally composed of three kinds of sensor: accelerometer, gyroscope and magnetometer. The calibration of these sensor suites not only requires turn-table or purpose-built fixture, but also entails a complex and laborious procedure in data sampling.

In this paper, we propose a method to calibrate a 9-DOF IMU by using a set of casually sampled raw sensor measurement. Our sampling procedure allows the sensor suite to move by hand and only requires about six minutes of fast and slow arbitrary rotations with intermittent pauses. It requires neither the specially-designed fixture and equipment, nor the strict sequences of sampling steps.

At the core of our method are the techniques of data filtering and a hierarchical scheme for calibration. All the raw sensor measurements are preprocessed by a series of band-pass filters before use. And our calibration scheme makes use of the gravity and the ambient magnetic field as references, and hierarchically calibrates the sensor model parameters towards the minimization of the mis-alignment, scaling and bias errors. Moreover, the calibration steps are formulated as a series of function optimization problems and are solved by an evolutionary algorithm. Finally, the performance of our method is experimentally evaluated. The results show that our method can effectively calibrate the sensor model parameters from one set of raw sensor measurement, and yield consistent calibration results.

I. INTRODUCTION

Motion sensing is vital to robotics [1] [2]. One of the workhorse sensing units in this field is the Inertial Measurement Unit (IMU). It combines the accelerometer, the gyroscope and sometimes the magnetometer to yield attitude and heading measurements. These measurements can facilitate motion control and trajectory planning. One of the key challenges in building these sensor suites is that, the raw measurements are usually coupled with large bias, non-unitary scaling and skew error between the axes. Without a proper identification of these sensor errors — a process known as the calibration of sensor, the IMU measurement can hardly be further processed to yield a consistent orientation or position estimations.

Although there are commercially available and factory-calibrated IMUs on the market which can provide accurate measurements, the high cost inhibits these units from volumetric use. For example, the number of required IMU proportionates with the number of agents in an unmanned aerial fleet [2]. Therefore, many [3]–[5] attempted to calibrate IMUs with the least possible equipment while without jeopardizing the measurement accuracy. However, these methods not only rely on precisely controlled calibration platforms, such as a 3-DOF rotational stages or a single axis turn-table, but also entail long and laborious procedures. Therefore, the calibration of an IMU

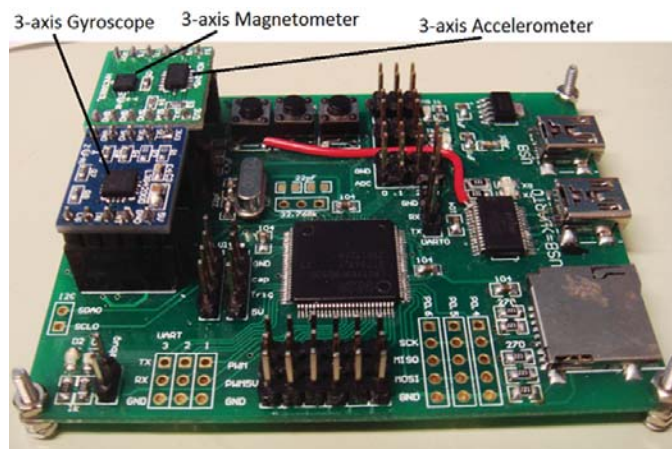


Fig. 1: Our sensor suite of the 9-DOF IMU that is composed of the gyroscope, the magnetometer and the accelerometer.

by making use of a set of samples that is freely collected by hand movement has been an open challenge.

Recently, two prominent methods [6], [7] have been proposed to calibrate an IMU without using special equipment. The key observation of their methods is that, the gravity vector measured by the accelerometers can act as a reference orientation, and guide the integration of the gyroscopic measurements. Straightforwardly, they use this insight to design a sensor error model and optimize the model parameters by using the collected samples. However, there are several limitations in these methods. Fong et. al. [6] focuses on a 6-DOF IMU that is composed of a tri-axial accelerometer and a tri-axial gyroscope and hence lack of references to calibrate the measurements on the yaw axis. On the other hand, Macias et. al. [7] only models the scaling factor and bias in the accelerometers. The errors in the gyroscope, such as the mis-alignment error, are neglected.

In this paper, we proposed a calibration method for a 9-DOF IMU. We not only consider the scaling factors, the axis mis-alignments and the bias in the sensor models for our sensor suite, but also take the magnetic distortions into account. Our method requires neither the external equipment nor the complex data sampling procedures. Our sampling procedure allows the sensor suite to move by hand and only requires about six minutes of fast and slow arbitrary rotations with intermittent pauses. It requires neither the specially-designed fixture and equipment, nor the strict sequences of sampling steps. Moreover, our calibration scheme makes use of the gravity and the ambient magnetic field as references, and hierarchically calibrates the sensor model parameters towards

the minimization of the mis-alignment, scaling and bias errors. The whole calibration is decomposed into a series of function minimization problems and are solved by an evolutionary algorithm. We test our method on a 9-DOF IMU and the results suggest that our method can effectively calibrate the sensor model parameters from one set of raw sensor measurement, and yield consistent calibration results.

This paper is organized as follows: Section II presents the sensors error models for the three tri-axial sensors and the cost functions to optimize the sensor error parameters. In Section III, we detail the design of a quasi-static detector and the data collection procedures. Section IV gives the steps of our hierarchical calibration algorithm where the optimization for each sensor is done by an evolutionary approach successively. In Section V, the calibration results for all the sensors are evaluated in experiments and are discussed individually. Finally, Section VI gives the conclusion.

II. SENSOR ERROR MODELS

The IMU used in this work is composed of three tri-axial MEMS sensors: ADXL345 accelerometer from Analog Devices. L3G4 gyroscope from STMicroelectronics and HMC5883L magnetometer from Honeywell. All three sensors output digital measurements, namely accelerations, angular rates and magnetic flux. A sensor error model and the corresponding calibration method are derived for each sensor.

A. Tri-axial Accelerometer

For the ease of communication, the parameters used in this paper follow the same naming conventions as in the traditional sensor error models [5] [6]. The general errors for the accelerometer measurements are the scaling errors, the axis mis-alignment, the static bias and the random noise. Therefore, the accelerometer error model is as followed,

$$\bar{\mathbf{a}} = \mathbf{K}_a \mathbf{T}_a \tilde{\mathbf{a}} - \mathbf{b}_a - \mathbf{n}_a \quad (1)$$

where $\tilde{\mathbf{a}}$ represents the raw accelerations measurements on each of its body x,y,z-axis and $\bar{\mathbf{a}}$ represents the calibrated accelerations with respect to the IMU frame. $\mathbf{K}_a$ is a 3x3 diagonal matrix containing three scaling factors in the accelerometer measurement on each axis. $\mathbf{b}_a$ is the static bias vector and $\mathbf{n}_a$ is a mean-zero Gaussian noise vector. The Gaussian noise such as the vibration noise on the mobile robots can be filtered by the band-pass filters, therefore it is not furthered discussed in this work. $\mathbf{T}_a$ is the axis mis-alignment correction matrix defined as followed,

$$\mathbf{T}_a = \begin{pmatrix} 1 & -\alpha_{yz} & \alpha_{zy} \\ \alpha_{xz} & 1 & -\alpha_{zx} \\ -\alpha_{xy} & \alpha_{yx} & 1 \end{pmatrix} \quad (2)$$

each α_{ij} represents the small angle deviation of the i-th sensitivity axis of the accelerometer around the j-th sensitivity axis of the IMU, as indicated in Fig. 2a. The mis-alignment error is usually small for the tri-axis MEMS sensors, but the mis-alignment between different tri-axial sensors can be large.

The calibration for the accelerometer aims to find the best fitted parameters in the constant matrices $\mathbf{K}_a \mathbf{T}_a$ and the constant vector $\mathbf{b}_a$ which constitute the least error in the acceleration measurement. Since the constant matrices $\mathbf{K}_a \mathbf{T}_a$

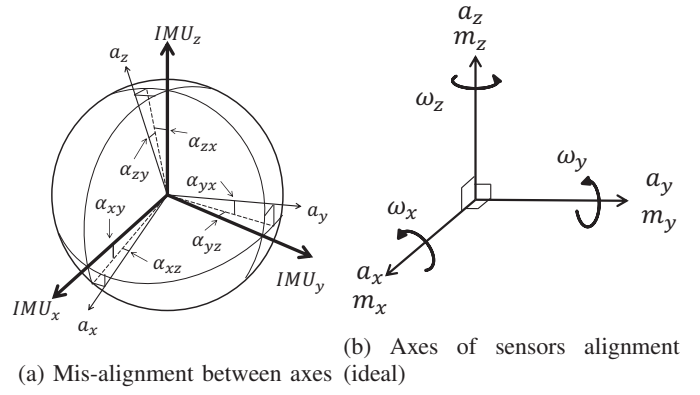


Fig. 2: Accelerometer, Magnetometer and Gyroscope sensors axes alignments and angular deviations.

will be multiplied together and become,

$$\Theta_a = \begin{pmatrix} \theta_{00} & \theta_{01} & \theta_{02} \\ \theta_{10} & \theta_{11} & \theta_{12} \\ \theta_{20} & \theta_{21} & \theta_{22} \end{pmatrix} \quad (3)$$

where Θ_a is a constant and diagonally dominant matrix, therefore the number of parameters being solved are reduced to twelve in total including nine elements in Θ_a and three elements in $\mathbf{b}_a$.

In the absence of the calibration equipment, the cost function, as proposed in [5], is utilizing the fact that the magnitude of gravity vector is $1g$. Therefore the cost is equal to the sum of squared errors of the N sets of static measurements from $1g$ such that the 12 parameters are optimized by,

$$(\Theta_a, \mathbf{b}_a) = \underset{\Theta_a, \mathbf{b}_a}{\operatorname{argmin}} \sum_{n=1}^N (1 - \|\bar{\mathbf{a}}\|_2)^2 \quad (4)$$

B. Tri-axial Magnetometer

Unlike the accelerometer measuring the constant gravity when it is placed statically, the magnetometer measures not only the constant earth magnetic field, but also the coupled magnetic noise induced from the nearby electronics and mechanical devices. Those noises are widely called soft-iron and hard-iron distortions [8] which is constant for a particular hardware settings. The traditional compensation schemes are adopted by using the fact that the magnitude of magnetic flux vector is constant at any orientations. The compensation is equivalent to transforming a center-shifted and rotationally scaled ellipse to a circle (or transforming a 3-D ellipsoid to a center-originated sphere). The sensor error model of magnetometer compensation is,

$$\bar{\mathbf{m}} = \mathbf{T}_m^{-1} \mathbf{K}_m \mathbf{T}_m \tilde{\mathbf{m}} - \mathbf{b}_m - \mathbf{n}_m \quad (5)$$

where $\tilde{\mathbf{m}}$ is the raw magnetometer measurement vector, $\bar{\mathbf{m}}$ is the compensated readings, $\mathbf{b}_m$ is the shifted ellipsoid center. Similar to the accelerometer model, $\mathbf{n}_m$ is the mean-zero noise vector which is also not tackled with in this paper. $\mathbf{K}_m$ is the scaling matrix along the major axes of the ellipse. Under the effect of soft-iron distortion, the scaling correction cannot be directly applied to the measurement vector $\tilde{\mathbf{m}}$, it should apply after the rotation transformation $\mathbf{T}_m$ which aligns the major axes onto the magnetometer frames.

Using the similarity between the magnetometer and the accelerometer which are both measuring vectors of constant magnitudes, the optimization cost function for accelerometer can also be applied to the magnetometer compensation. Although the magnitude of the magnetic flux is not necessarily unitary anywhere, it is reasonable to set the ideal magnetic flux magnitude to 1 unit and apply to the cost function because in usual practice only the direction information of the magnetic flux vector is used for heading referencing and the norm information is usually neglected. Hence the optimized parameter set $(\mathbf{T}_m, \mathbf{K}_m, \mathbf{b}_m)$, the cost function of the magnetometer error model is,

$$\mathbf{L}(\mathbf{T}_m, \mathbf{K}_m, \mathbf{b}_m) = \sum_{n=1}^N (1 - \|\tilde{\mathbf{m}}\|_2)^2 \quad (6)$$

C. Tri-axial Gyroscope

Similar to the accelerometer error model, the calibration of the gyroscope also includes the diagonal scaling matrix $\mathbf{K}_g$, the axis mis-alignment matrix $\mathbf{T}_g$, the bias vector $\mathbf{b}_g$ and the mean-zero noise vector $\mathbf{n}_g$. The gyroscope error model is given by,

$$\bar{\mathbf{g}} = \mathbf{K}_g \mathbf{T}_g \tilde{\mathbf{g}} - \mathbf{b}_g - \mathbf{n}_g \quad (7)$$

where $\bar{\mathbf{g}}$ is the corrected angular rates measurement in radian per second and $\tilde{\mathbf{g}}$ is the raw measurement vector with respect to the sensor axes. And this error model can also be simplified one step further to 12 calibration parameters including Θ_g and $\mathbf{b}_g$ where Θ_g is the resultant matrix multiplied by the mis-alignment matrix, the scaling matrix and the rotation matrix that aligns the gyroscope sensor frame on the calibrated accelerometer frame.

Being different from the cases of accelerometer and magnetometer calibrations, since there is no accurate platform measuring precise angular velocities, direct comparisons between ideal and noisy angular rates no longer applies. However, the cost function proposed in [6] for the gyroscope calibration provides a suitable means to optimize the gyroscope parameters, it basically calculates the resultant orientation change between 2 static intervals by integrating the measured angular rates, then applies the resultant rotation to the prior static calibrated gravity vector to yield the new gravity vector. By adopting the fact that ideally the rotation by angular rates integration should result in the same orientation change reflected by the differences between the prior and posterior gravity vector, the errors between the gravity vector in the posterior static interval and the new gravity vector serves as the cost function for calibrating the rate gyroscope. This approach needs to restrict the resultant as well as the incremental rotations to be about a horizontal axis, otherwise the yaw errors without gravity reference will be propagating to gyroscope error models. A modified cost function is proposed here by taking the calibrated magnetometer measurement vector into account to eliminate the yaw error without the aforementioned limitation during the data collection process.

In the gyroscope calibration, the quaternions for rotation is used instead of Euler representation to avoid the gimbal-lock singularity [9]. Between 2 static intervals, the next quaternion at time $t+1$, $\mathbf{q}'$ is updated by current quaternion $\mathbf{q}$ and the incremental quaternion change $\delta\mathbf{q}$. The governing equation is

given by,

$$\begin{pmatrix} q0' \\ q1' \\ q2' \\ q3' \end{pmatrix} = \begin{pmatrix} q0 \\ q1 \\ q2 \\ q3 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -g_x q1 - g_y q2 - g_z q3 \\ g_x q0 + g_z q2 - g_y q3 \\ g_y q0 - g_z q1 + g_x q3 \\ g_z q0 + g_y q1 - g_x q2 \end{pmatrix} \quad (8)$$

where the angular rates measurement $\bar{\mathbf{g}} = (g_x, g_y, g_z)^T$ is computed from equation (7).

The resultant orientation $\mathbf{q}_T = (w, x, y, z)^T$ after integration between two static intervals is used to get the rotation matrix $\mathbf{R}$ which is defined as,

$$\mathbf{R} = \begin{pmatrix} w^2 + x^2 - y^2 - z^2 & 2xy - 2wz & 2xz + 2wy \\ 2xy + 2wz & w^2 - x^2 + y^2 - z^2 & 2yz - 2wx \\ 2xz - 2wy & 2yz + 2wx & w^2 - x^2 - y^2 + z^2 \end{pmatrix} \quad (9)$$

The equation for computing the new gravity and magnetic flux vectors $\mathbf{v}_a$ and $\mathbf{v}_m$ initiating from the prior calibrated gravity and magnetic flux vectors $\mathbf{u}_a$ and $\mathbf{u}_m$ using the above rotation matrix is then given by,

$$(\mathbf{v}_a, \mathbf{v}_m) = \mathbf{R}(\mathbf{u}_a, \mathbf{u}_m) \quad (10)$$

Now the cost function for calibrating the rate gyroscope is modified such that the parameters to be optimized are,

$$(\Theta_g, \mathbf{b}_g) = \underset{\Theta_g, \mathbf{b}_g}{\operatorname{argmin}} \sum_{n=1}^N (\|\mathbf{v}_a - \mathbf{u}'_a\|_2 + \|\mathbf{v}_m - \mathbf{u}'_m\|_2) \quad (11)$$

where $\mathbf{u}'_a$ and $\mathbf{u}'_m$ are the calibrated gravity and magnetic flux vectors in the posterior static interval by the accelerometer and magnetometer. The above optimization for the gyroscope is valid for arbitrary motions between 2 static intervals. The calibrated vectors $\mathbf{u}_a, \mathbf{u}_m, \mathbf{u}'_a, \mathbf{u}'_m$ are calculated by applying the optimized parameter sets after the accelerometer and the magnetometer calibration to their raw measurements.

III. CALIBRATION EXPERIMENT PROCEDURE

At the absence of a precisely controlled platform, calibration of accelerometer and compensation of magnetometer can only utilize the fact that the magnitudes of the gravity vector and that of magnetic vector are constants, while the gyroscope calibration adopts the property that ideal rotation by integrating the angular rates is identical to the orientation changes of the calibrated reference gravity and magnetic vectors.

The calibrations of the three sensors are based on the measurements of different characteristics. The accelerometer calibration requires static measurements at different orientations, and the magnetometer needs the measurements under the quasi-static motion¹ in order to better spread the magnetic flux measurements on the surface of an ellipsoid. Additionally, the gyroscope needs to be excited by rotational motions ranging from small to large angular speeds for better calibration result on the scaling factors and bias. Even though the data collection can be done separately, some detection methods are still necessary to verify whether the IMU is under the quasi-static states or in rapid motions which causes large errors but invisible to human. The quasi-static detector proposed in [6] [10] are modified to incorporate the detection of magnetic flux

¹ the quasi-static motion here refers to the slow rotations (about 10 seconds per revolution) that are closed to the equilibrium states.

changes in order to get rid of cases that the IMU is static but the magnetic field change due to the environment because that can cause significant errors in the magnetometer calibration. We denote the symbols $LPF()$, $RECT()$, $HPF()$ as the low-pass filter, the rectification operator and high-pass filter respectively. the quasi-static detector s_a for accelerometer is,

$$s_a = \|LPF(RECT(HPF(\tilde{a})))\|_2 \quad (12)$$

where $\tilde{a}$ is the raw acceleration measurements. Similar quasi-static detectors for gyroscope and magnetometers are defined as,

$$s_m = \|LPF(RECT(HPF(\tilde{m})))\|_2 \quad (13)$$

$$s_g = \|LPF(RECT(\tilde{g}))\|_2 \quad (14)$$

Since the rate gyroscope outputs angular rates so no $HPF()$ is needed. To better combined these terms, a quasi-static coefficient is introduced as followed,

$$s = 1/(1 + s_a + s_m + s_g), \quad \text{where } s \in (0, 1] \quad (15)$$

If the value of s is closer to 1, it reveals the state of that measurement is closer to the static state and vice versa.

In the calibration experiments, measurements having a threshold value of greater than 0.98 of s is used for the accelerometer and magnetometer calibrations and the others measurements are used for the gyroscope calibrations. Increasing the quasi-static coefficient threshold will generally improve the calibration results of the accelerometer and the magnetometer. However, setting this value too high may cause large reduction on the number of measurements datasets and thus reducing the accuracy of the calibrations. The effects of varying this coefficient from 0.90 to 0.99 at a step size of 0.1 is manually tested and the calibrations results of the accelerometer and magnetometer show that the least variance of errors is achieved at $s = 0.98$. In spite of this rough finding in this paper, deeper analysis on the effect of the calibration performance by varying the quasi-static coefficient threshold will be left for future discussion.

Nevertheless, with the help of the quasi-static coefficient, different sensors calibration stages will search for the interested measurement datasets, therefore the data collection procedures can be simplified to just one set of measurements coupled with various rotational motions and static placements at different pauses. Examples of the static placements of different orientations for calibrating the accelerometer are given by Fig. (3), and examples of the suggested rotational movements are shown in Fig. (4).

Since the number of parameters of the accelerometer sensor error model is 12, the number of independent datasets should be at least 12 to ensure the cost function will result in a solution which is not under-determined. The suggested static pauses given in Fig. (3) will give a total of 18 sets of independent measurements which spreads more or less on the entire surface of a distorted sphere. Similar explanations apply to the rotational motions for the gyroscope and the magnetometer calibrations. Positive and negative rotations about x-,y-,z-axis will output a total of 6 independent datasets, while there is another 12 measurement sets rotating in both directions about 6 cross-axes on the three principle planes, where Fig. (4b) is an example showing the 2 cross-axes on the x-y plane, and

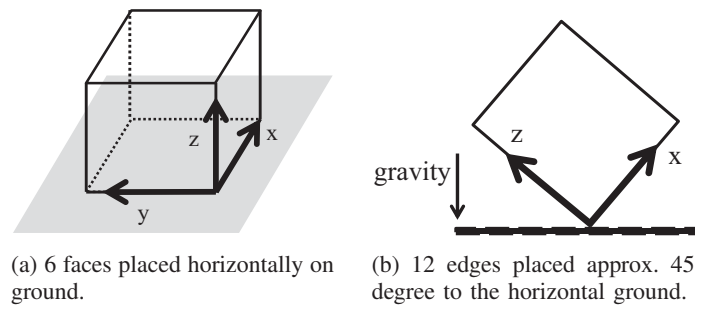


Fig. 3: Static placements for accelerometer calibrations.

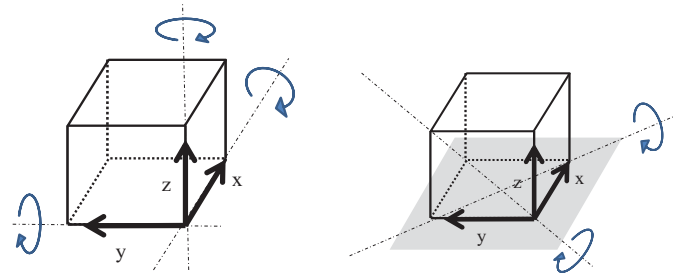


Fig. 4: Rotational motions for the gyroscope calibrations.

the other 4 cross-axes are lying on the y-z plane and x-z plane respectively. It should be noted that since the magnetometer needs more quasi-static measurements on magnetic flux than the accelerometer does in order to have the datasets that spreads more evenly and intensively the surface of a center-shifted and rotationally scaled ellipsoid, general and slow rotations are required.

A portion of the raw measurements from 3 sensors is shown in Fig. (5). The measurements are taken from an integrated IMU at a rate of 100Hz. The unit of acceleration measurements is g where $1g = 9.81ms^{-2}$. The unit of the gyroscope data is rad/s and that of magnetometer measurements is $milligauss$. The value of the quasi-static coefficient in the 4th row indicates how static the measurement is. States having a larger value of the Quasi-static coefficient represent more stable states. It can be seen that the earlier periods include coarser and slower motions with longer steady pauses, those states are having a value of quasi-static coefficient closed to 1, while later stages with large drops of the coefficient are much vigorous rotations aiming to excite the range of the gyroscope.

IV. CALIBRATION ALGORITHM

The full calibration flow of the proposed calibration is illustrated in Algorithm 1. The proposed calibration procedures realize a self-calibration method of a 9-DOF IMU which uses the constant gravity and the magnetic flux as reference vectors and further calibrate the gyroscope without the propagation of yaw errors as in the traditional 6-DOF IMU calibrations. Therefore the IMU can be calibrated anytime by itself without any external equipment in the operating zone.

The calibrations in line 4,5 and 16 of Algorithm 1 use an evolutionary direct searching algorithm aiming to get rid

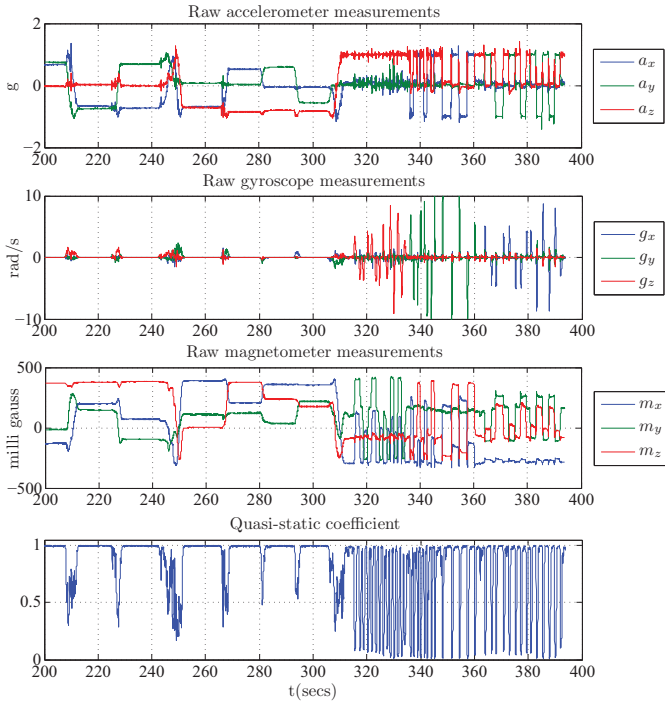


Fig. 5: A portion of the experiment data set

Algorithm 1 IMU Calibration By One Training Dataset

- 1: $(\tilde{\mathbf{a}}, \tilde{\mathbf{m}}, \tilde{\mathbf{g}}) = \text{GetRawMeasurement}()$
 - 2: Determine quasi-static coefficients s_i for the raw datasets where $i \in [1, \text{datalength}]$
 - 3: $\forall \tilde{\mathbf{a}}_i \in \tilde{\mathbf{a}}$ and $\tilde{\mathbf{m}}_i \in \tilde{\mathbf{m}}$, where $s_i > 0.98$
 - 4: Calibrate the accelerometer $(\Theta_a, \mathbf{b}_a)$ by eqt. (4)
 - 5: Calibrate the magnetometer $(\mathbf{T}_m, \mathbf{K}_m, \mathbf{b}_m)$ by eqt. (6)
 - 6: $\bar{\mathbf{a}} = \text{GetCalibratedGravity}(\Theta_a, \mathbf{b}_a)$
 - 7: $\bar{\mathbf{m}} = \text{GetCalibratedMagneticFlux}(\mathbf{T}_m, \mathbf{K}_m, \mathbf{b}_m)$
 - 8: Split $\bar{\mathbf{a}}, \bar{\mathbf{m}}, \tilde{\mathbf{g}}$ to K segments by a cut-off value $s_i = 0.98$
 - 9: For each segment of period T having $s_i < 0.98$,
 - 10: $(\mathbf{u}_a, \mathbf{u}_m) = \text{GetPriorReferenceVectors}(\bar{\mathbf{a}}, \bar{\mathbf{m}})$
 - 11: $(\mathbf{u}'_a, \mathbf{u}'_m) = \text{GetPosteriorReferenceVectors}(\bar{\mathbf{a}}, \bar{\mathbf{m}})$
 - 12: $\mathbf{q}_0 = \text{InitializeQuaternion}(1, 0, 0, 0)$
 - 13: $\mathbf{q}_T = \int_0^T \mathbf{q} d\mathbf{q}$ in eqt. (8)
 - 14: $\mathbf{R} = \text{GetRotationMatrix}(\mathbf{q}_T)$ by eqt. (9)
 - 15: $(\mathbf{v}_a, \mathbf{v}_m) = \text{GetNewVectors}(\mathbf{R}, \mathbf{u}_a, \mathbf{u}_m)$
 - 16: Calibrate the gyroscope $(\Theta_g, \mathbf{b}_g)$ by eqt. (11)
 - 17: Return parameters $(\Theta_a, \mathbf{b}_a, \mathbf{T}_m, \mathbf{K}_m, \mathbf{b}_m, \Theta_g, \mathbf{b}_g)$
-

of trapping in the local minima and the mis-convergence due to settings of the initialization parameters. Among so many evolutionary approaches, the differential evolution is selected due to its particular strength on tackling real-valued optimization problems [11], [12] in the divergence of the search spaces and the fast convergence time.

V. CALIBRATION RESULTS

Since one set of measurements can already calibrate the three sensors hierarchically, twelve raw datasets of approximately six minutes are taken for the calibrations. All the

TABLE I: Mean and standard deviation of the parameters for the accelerometer and the gyroscope.

Parameters	Accelerometer		Gyroscope	
	Mean	S.D.	Mean	S.D.
θ_{00}	0.9828	0.0005	0.9911	0.0030
θ_{01}	0.0131	0.0278	0.0050	0.0135
θ_{02}	0.0031	0.0228	0.0073	0.0097
θ_{10}	-0.109	0.0278	-0.0090	0.0159
θ_{11}	0.9825	0.0010	0.9456	0.0023
θ_{12}	-0.0038	0.0346	0.0211	0.0215
θ_{20}	-0.0118	0.0226	-0.0102	0.0075
θ_{21}	0.0032	0.0338	-0.0205	0.0174
θ_{22}	0.9983	0.0007	1.0140	0.0064
b_x	0.0693	0.0008	-0.0016	0.0007
b_y	0.0213	0.0006	-0.0090	0.0016
b_z	0.0208	0.0014	0.0103	0.0018

datasets follow the suggested motional profiles in Fig. (3) and Fig. (4). Table I shows the calibration results on the parameters for both the accelerometer and the gyroscope since they share the same number of model parameters. θ_{ij} , which are dimensionless, are the elements in the resultant matrices (Θ_a, Θ_g) combined with the scaling and mis-alignment matrix. The units of the bias vectors $\mathbf{b}$ is g for the accelerometer and rad/s for the gyroscope. It can be seen from Table I that if the mis-alignment factors are small (mean values of un-highlighted rows), the most effective factors determining the resulting errors in the error models are the scaling factors and the bias. The small standard deviations on the scaling factors and the bias rows show the consistence of those parameters of our methods.

From Table II, the aforementioned explanation also applies to the consistent convergence on the scaling coefficients as well as the bias, which is in the unit of *milligauss*, for the magnetometer. The experimental data collections were done indoor using the configuration shown in Fig. (1) which is not accompanied with any metallic structures, hence the effect of soft-iron distortion is small as revealed from the three almost-unitary scaling factors, this makes the $\mathbf{T}_m$ matrix in eqt. (5) ill-defined and thus reasonably resulting in large variations on the elements t_{ij} in $\mathbf{T}_m$.

The selected parameter sets for the accelerometer and gyroscope error models are set according to the mean values from Table I. Due to the large variances of the elements in matrix $\mathbf{T}_m$, the best parameter vector which constitutes the least error value of the cost function (6) is selected instead of feeding the irrational average values to the magnetometer error models.

To evaluate the performance of the calibrated sensors, the approach in Fong et. al. [6] is used. The errors that are determined by the norm-differences between the raw measurement vectors and the calibrated vectors, are represented by the maximum angular errors, such that the error of the gravity measurements measured from the accelerometer, $\delta = \text{abs}(\|\mathbf{a}\|_2 - 1)$ and the maximum angular error is defined by $\sin^{-1}(\delta)$. It assumes the error portion entirely comes from the horizontal component perpendicular to the gravity. The analysis of the sensor's performance is done using 12 sets of extra experiments with the same settings of the quasi-static

TABLE II: Mean and standard deviation of the parameters for the magnetometer.

Parameters	Magnetometer		
	Mean	S.D.	Best set
t_{00}	-0.998	0.6275	0.8629
t_{01}	0.2754	0.6559	0.4802
t_{02}	0.0435	0.2973	0.1579
t_{10}	-0.0282	0.6236	0.1681
t_{11}	0.1024	0.5418	0.0219
t_{12}	-0.0762	0.5482	-0.9855
t_{20}	-0.1071	0.4509	-0.4767
t_{21}	0.2107	0.3843	0.8769
t_{22}	0.4752	0.6061	-0.0618
s_x	0.0021	0.0001	0.0021
s_y	0.0021	0.0001	0.0022
s_z	0.0022	0.0001	0.0020
b_x	35.5951	1.6894	34.0730
b_y	79.4491	1.3585	78.3904
b_z	71.8808	0.9190	71.7636

TABLE III: Mean, standard deviation and max. observed errors in magnitudes (angular error).

	Accelerometer errors (mg)		
	Mean error	S.D. of error	Max error
Raw	37.7 (2.16°)	22.8(1.36°)	104.3(5.99°)
Calibrated	3.4 (0.19°)	3.0(0.17°)	43.8(2.5°)

TABLE IV: Mean, standard deviation and max. observed errors in magnitudes from ideal 1 unit of magnetic flux.

	Magnetometer errors (milli gauss)		
	Mean error	S.D. of error	Max error
Raw	58.6 (3.36°)	42.5(2.46°)	131.4(7.55°)
Calibrated	9.2 (0.53°)	9.2(0.53°)	84.5(4.85°)

threshold in algorithm 1. Table III displays the differences of performance before and after calibrations of the accelerometer. The large reductions in the average errors, standard deviations as well as the maximum errors in norm of the gravity verify that the sensors errors have been greatly suppressed.

Similarly the analysis of the calibrated magnetometer utilized the constant property of the magnitude of the magnetic flux within a small region. Table IV shows similar improvements for the magnetometer after calibrations. The variance and the average error are both reduced by more than a factor of 5, this proves the calibrated magnetometer performs much more consistent and closed to the ideal sensor model.

For the final analysis of the calibrated gyroscope in Table V, the calibrated accelerometer and magnetometer measurements need to be computed first to ensure the evaluation focusing only on the improvement on the performance of the calibrated gyroscope. From procedures 8-15 in Algorithm. 1, the necessary vectors are computed and the errors of eqt. (11) are modified by spitting the error of gravity vector difference and the error of magnetic flux vector difference for display reasoning. Since the experiment is done by general hand-held rotations mixed with slow and fast motions following basically the aforesaid calibration motions in Fig. (3) and (4), some

TABLE V: Errors in magnitude of the gravity vectors difference (mg) and magnetic flux vectors difference ($milligauss$) in eqt. (11). before and after the gyroscope calibration.

	Gravity Difference		Magnetic Flux Difference	
	Before	After	Before	After
Mean error	68.4(3.92°)	30.8(1.76°)	53.9(3.08°)	24.6(1.41°)
S.D.	47.3(2.71°)	20.2(1.16°)	37.7(2.16°)	16.3(0.93°)
Max. error	317.2(18.49°)	130.5(7.50°)	250.9(14.53°)	103.2(5.92°)

imperceptible pauses may sometime spit large rotations into many data segments with a very short period of time, this leads to an increasing number of small gravity and magnetic flux vectors errors and results in small average errors. However, the calibration results suggest that the mean errors and the standard deviations are at least reduced by a factor of 2 after the gyroscope calibrations while the maximum errors, which comes from fast revolutionary motions about $360^\circ/s$, are also reduced to a value less than 8 degrees.

VI. CONCLUSION AND FUTURE WORK

This paper presents a calibration method for a 9-DOF IMU that requires neither the specially-designed equipments nor the long and laborious procedures for data sampling. Our procedure of the data sampling allows the sensor suite to move by hand, and only requires a few minutes of arbitrary rotations with intermittent pauses. And then the calibration results is automatically generated by an evolutionary algorithm in less than an hour. Looking forward, we will examine the performance of the proposed method under different acceleration profiles, different rotation movements and speeds in the near future.

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